



**Bahir Dar University**  
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**Advanced Applied Mathematics III (Math 2042) worksheet II**

1. Find the arc length, the unit tangent vector, the unit normal and the curvature of a curve on the given interval whose parameterization is given below
  - a)  $r(t) = (1 - 2\cos t)i + 2\sin t j - 12tk$ ;  $0 \leq t \leq 1$
  - b)  $r(t) = \cos^3 t i + \sin^3 t j + 3k$ ;  $0 \leq t \leq \frac{\pi}{2}$
2. Determine the curl and divergence of  $F(x, y, z) = e^{-xy}i + xe^{-yz}j + xe^{-xz}k$
3. Determine whether F conservative or not and find the potential function if F conservative.
  - a)  $F(x, y) = ye^{xy}i + (xe^{xy} + \cos y)j$       c)  $F(x, y, z) = (2x + 3)i + (2y + z)j + yk$
  - b)  $F(x, y, z) = 4xe^z i + \cos y j + 2x^2 e^z k$
4. Evaluate the following line integrals
  - a)  $\int_C (y + 2xe^y) dx + (x + x^2 e^y) dy$ , where  $C: r(t) = \sqrt{t}i + \ln t j + tk$ , for  $1 \leq t \leq 4$
  - b)  $\int_C (x + y) ds$ , where  $C: r(t) = (e^t + 1)i + (e^t - 1)j$ , for  $0 \leq t \leq \ln 2$
  - c)  $\int_C (2xy - 5yz) ds$  where  $C$  is the line segment from  $(0, 1, 0)$  to  $(0, 3, 2)$ .
  - d)  $\int_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$  where  $C$  is a circle of radius 2
5. Evaluate  $\int_C F \cdot dr$  where;
  - a)  $F(x, y, z) = zi - yj - xk$  and  $C: r(t) = 5i - \sin t j - \cos t k$ , for  $0 \leq t \leq \frac{\pi}{4}$
  - b)  $F(x, y, z) = ye^{x^2} i + x \cos y^3 e^{y^2} k$  and  $C: r(t) = e^t i + e^t j + 3k$ , for  $0 \leq t \leq 1$
  - c)  $F(x, y, z) = \frac{8x^3}{1+x^4+y^4+z^4} i + \frac{8y^3}{1+x^4+y^4+z^4} j + \frac{8z^3}{1+x^4+y^4+z^4} k$  and  $C: r(t) = t^2 i + t^4 j + t^6 k$ , for  $0 \leq t \leq 2$ .
6. Use Green's theorem to evaluate (C is oriented positively)
  - a)  $\int_C 2x^3 y dx - 3x^2 y^2 dy$ ; where  $C$  is a triangle with vertices  $(0, 1)$ ,  $(3, 4)$  &  $(2, 7)$
  - b)  $\oint_C y \sin x dx - \cos x dy$ ; where  $C$  is the path from  $(3, 0)$  to  $(-3, 0)$  along the curve  $y = \sqrt{9 - x^2}$  and from  $(-3, 0)$  to  $(3, 0)$  along the  $x$ -axis.
  - c)  $\oint_C (y - x) dx + (2x - y) dy$ ; where  $C$  is the boundary of the region between the graphs of  $y = x$  and  $y = x^2 - 2x$ .
  - d)  $\oint_C e^{e^x} - 2y dx + \tan^{-1}(ye^{-y} + 1) dy$ , where  $C$  is the ellipse  $9x^2 + 4y^2 = 36$ .
7. Evaluate the following surface integrals
  - a)  $\iint_S \sqrt{4x^2 + 4y^2 + 1} dS$ , where  $S$  is part of the paraboloid  $z = x^2 + y^2$  below the plane  $y = z$
  - b)  $\iint_S 5zy^2 dS$ , where  $S$  is the portion of the cone  $z^2 = 2x^2 + 2y^2$  that lies between the planes  $z = 1$  and  $z = 2$ .

- c)  $\iint_S z(x^2 + y^2) dS$ , where  $S$  is the hemisphere  $\sqrt{x^2 + y^2 + z^2} = 2$ .
- d)  $\iint_S \sqrt{1 + 4z} dS$ , where  $S$  is part of the paraboloid  $z = x^2 + y^2$  inside the cylinder  $x^2 + y^2 = 2$ .

8. Evaluate  $\iint_S F \cdot n dS$ , where

- a)  $F(x, y, z) = xi - j + 2x^2k$  and  $S$  is part of the paraboloid  $z = x^2 + y^2$  above the  $xy$  plane bounded by the parabolas  $x = 1 - y^2$  and  $x = y^2 - 1$ .
- b)  $F(x, y, z) = y(x^2 + y^2)^{\frac{3}{2}}i + x(x^2 + y^2)^{\frac{3}{2}}j + (z + 1)k$  and  $S$  the boundary of the solid region bounded above by the plane  $z = 2x$  and below by the paraboloid  $z = x^2 + y^2$ .
- c)  $F(x, y, z) = (3x^4 + e^{y^3})i + (x^3 + y^3)j + 12xy^2zk$  and  $S$  is the surface bounded by the cylinder  $x^2 + y^2 = 1$  and the planes  $z = 6x + 5$  and  $z = 0$ .

9. Evaluate  $\iint_S (\text{curl} F) \cdot n dS$ , where  $F(x, y, z) = x^3yi - y^3j + z \sec(xyz)k$  and  $S$  is the portion of the sphere  $\rho = \sqrt{2}$ , above the plane  $z = 1$ , with outward normal.

10. Use Stokes' Theorem to evaluate :

- a)  $\int_C F \cdot dr$ , where  $F(x, y, z) = (x^2 + z)i + (y^2 + x)j + (z^2 + y)k$  and  $C$  is the intersection of the sphere  $\rho = 1$  and the cone  $z = \sqrt{x^2 + y^2}$  with outward normal.
- b)  $\oint_C F \cdot dr$ , where  $F(x, y, z) = 2yi + 3zj - 2xk$  and  $C$  is part of the sphere  $\rho = 1$  in the first octant with outward normal.
- c)  $\oint_C F \cdot dr$ , where  $F(x, y, z) = z^2i + xj - 3zk$  and  $C$  is the intersection of the plane  $z + 2y = 6$  and the cylinder  $x^2 + y^2 = 4$  with outward normal.